

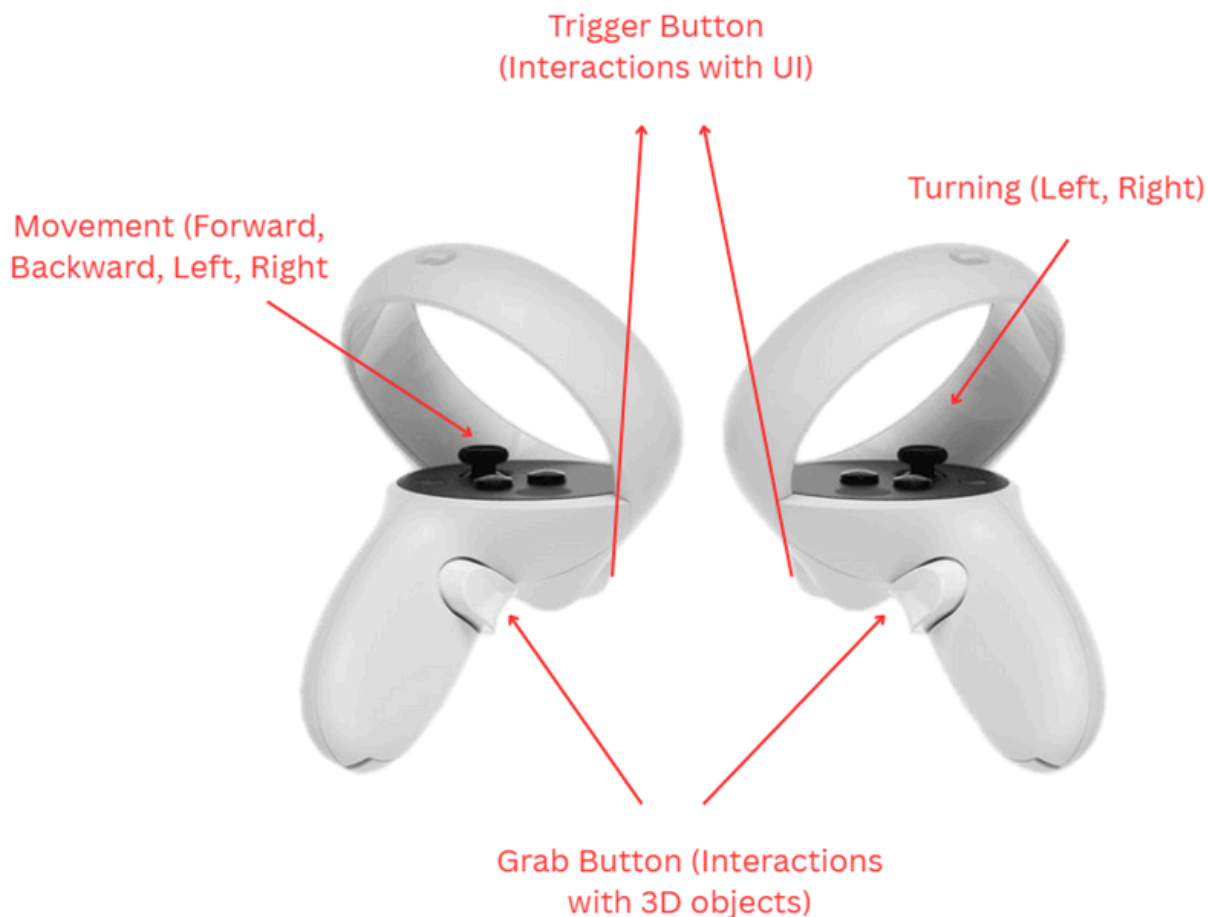
World Without Assumptions

Title & Team Info

- **Game Title:** World Without Assumptions
- **Team Name:** Aoe Studios
- **Team Members:** Joel Wong, Javier Leo, Rasuil, Zenon
- **Platform:** Oculus Quest 2

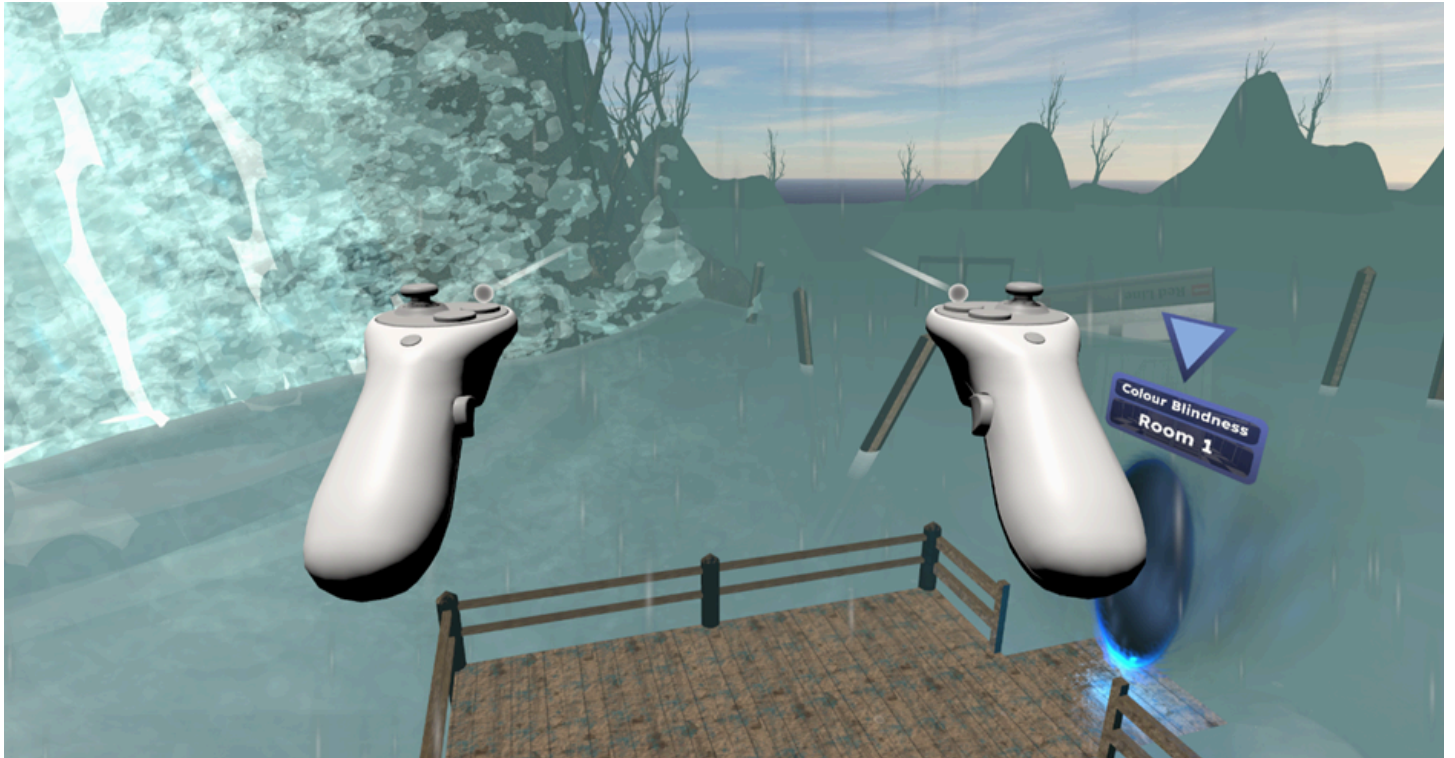
Controls:

Game Controls: Movement Joystick for moving front, back, left and right, Turning Joystick for rotating left and right, Grab Buttons to Interact with 3D objects, Trigger button to interact with UI.



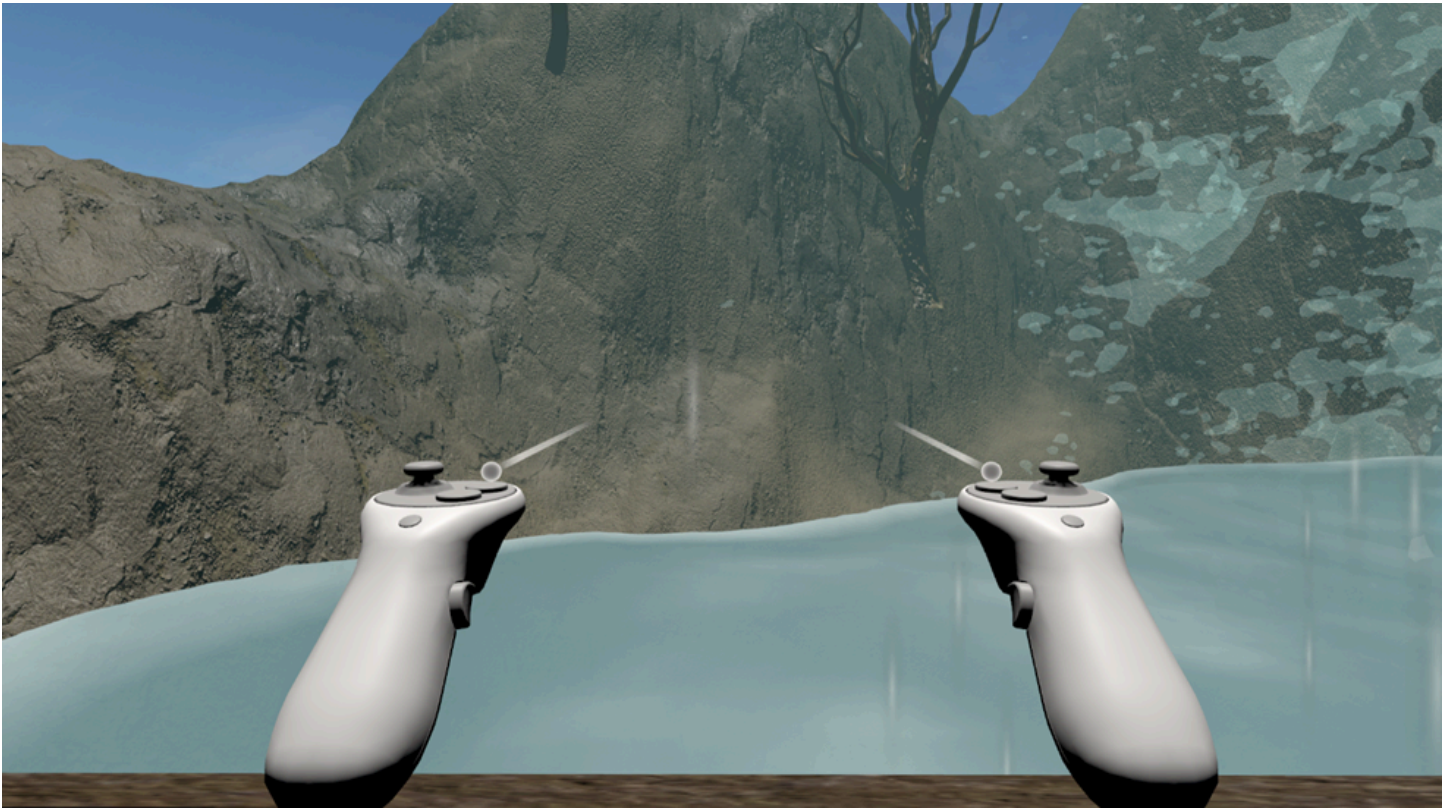
Limitations/Bugs:

Gravity: Player going up using the A button but not dropping down despite gravity being turned on





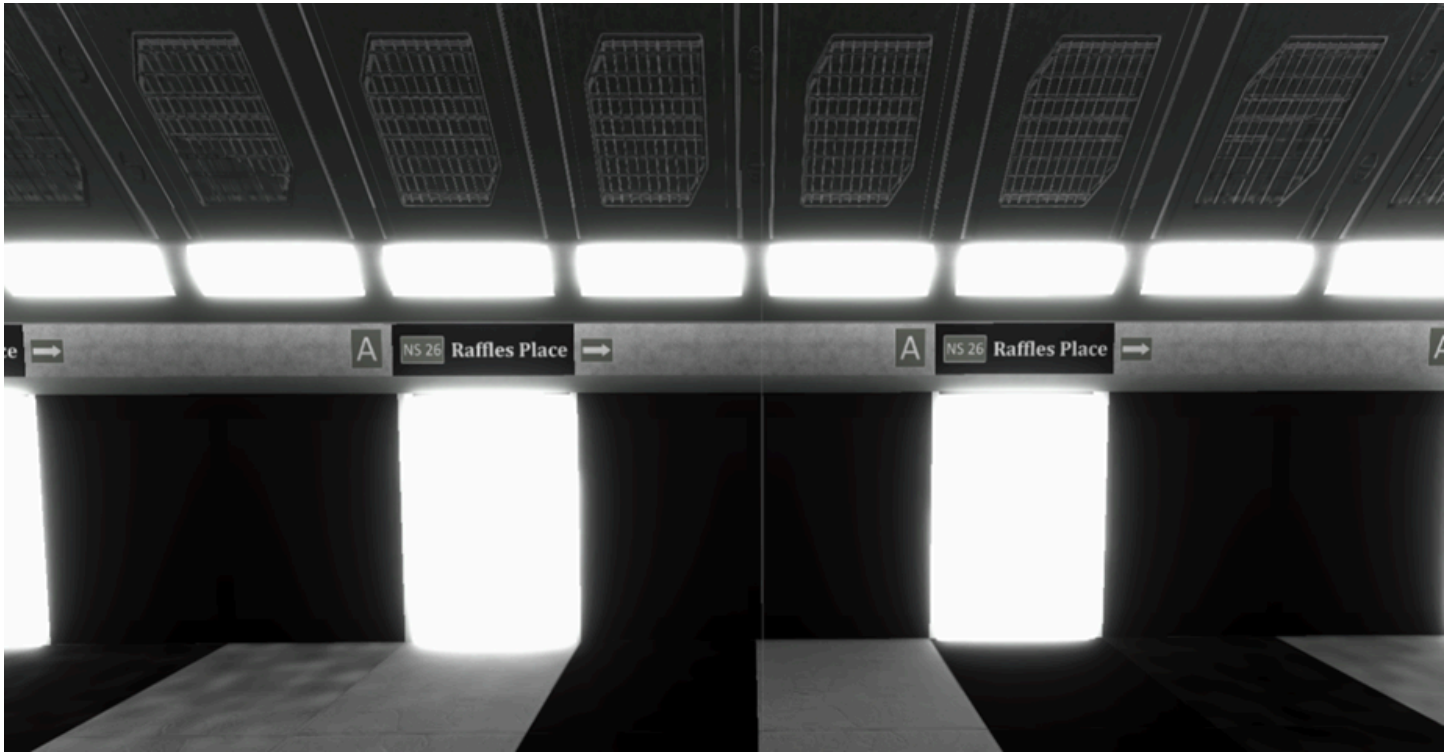
Player Spawning wrong direction: If the player starts out facing the front when using the headset, but turns around and open the game, the game will spawn player facing backwards.



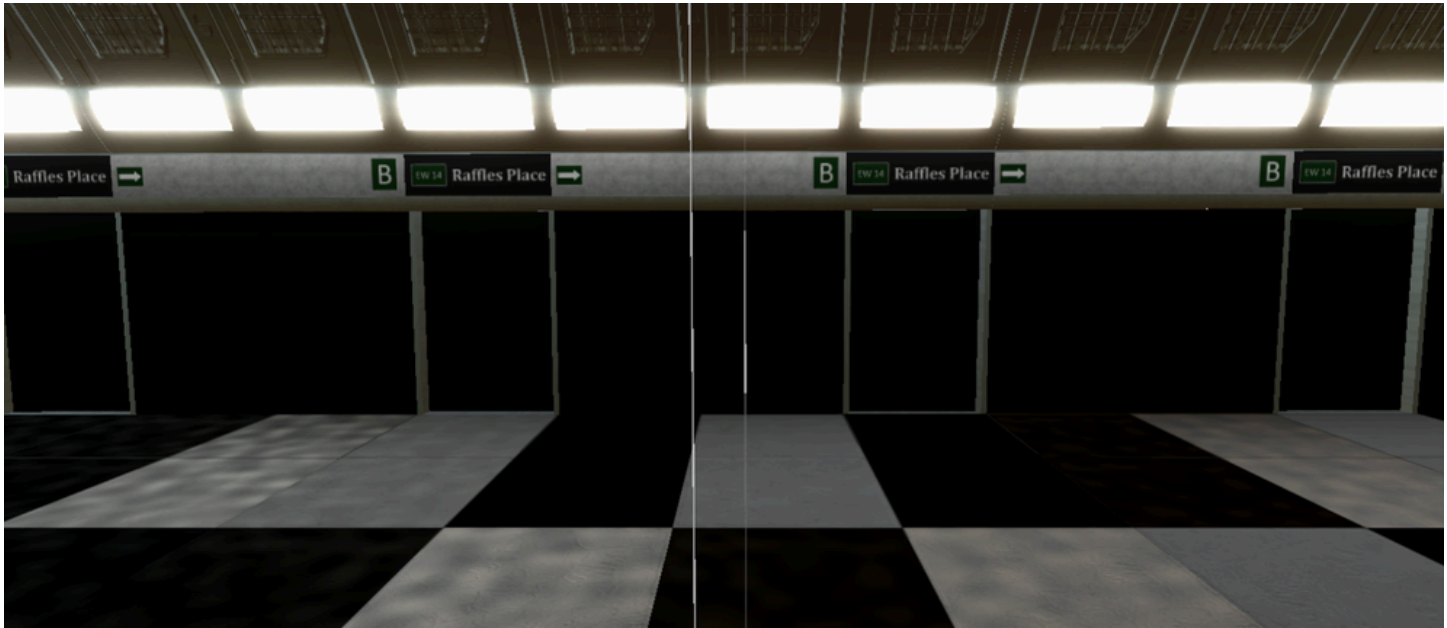
Player heads phasing: If player moves their whole head with the headset outside the room they may glitch through the walls even with box collider blocking them, which is a general VR headset issue.

Puzzles:

Room 1 Build : The right or wrong would not matter as player will only be told whether they are right or wrong but it won't affect the gameplay, but the platform with the NS on the sign will be the right platform to enter.



Room 1 Redesign: For the redesign stage the two things to redesign are the Map and the platform.



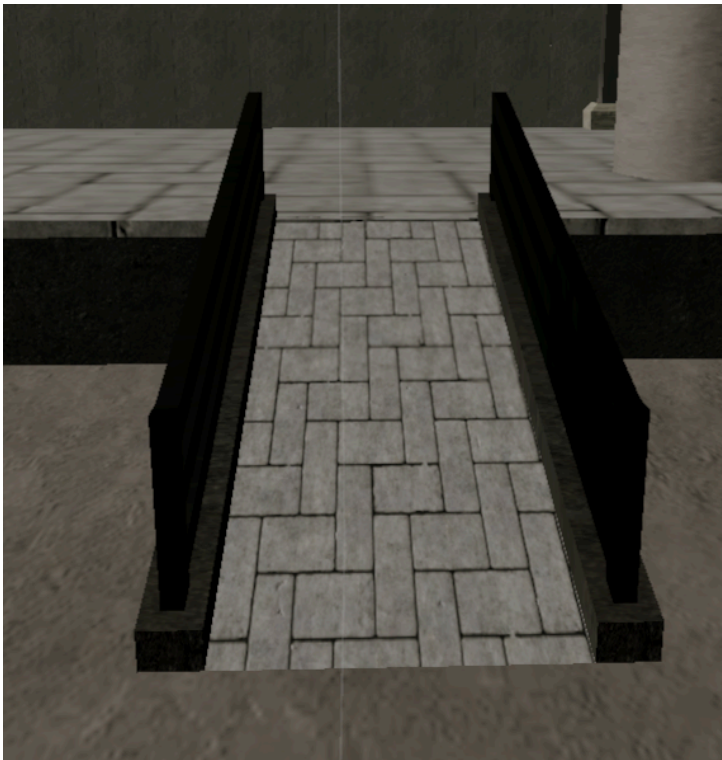


Room 2 Redesign: There isn't any right or wrong in the other rooms, but for the redesign in room 2 the 2 things to redesign are the Lights and the doors.



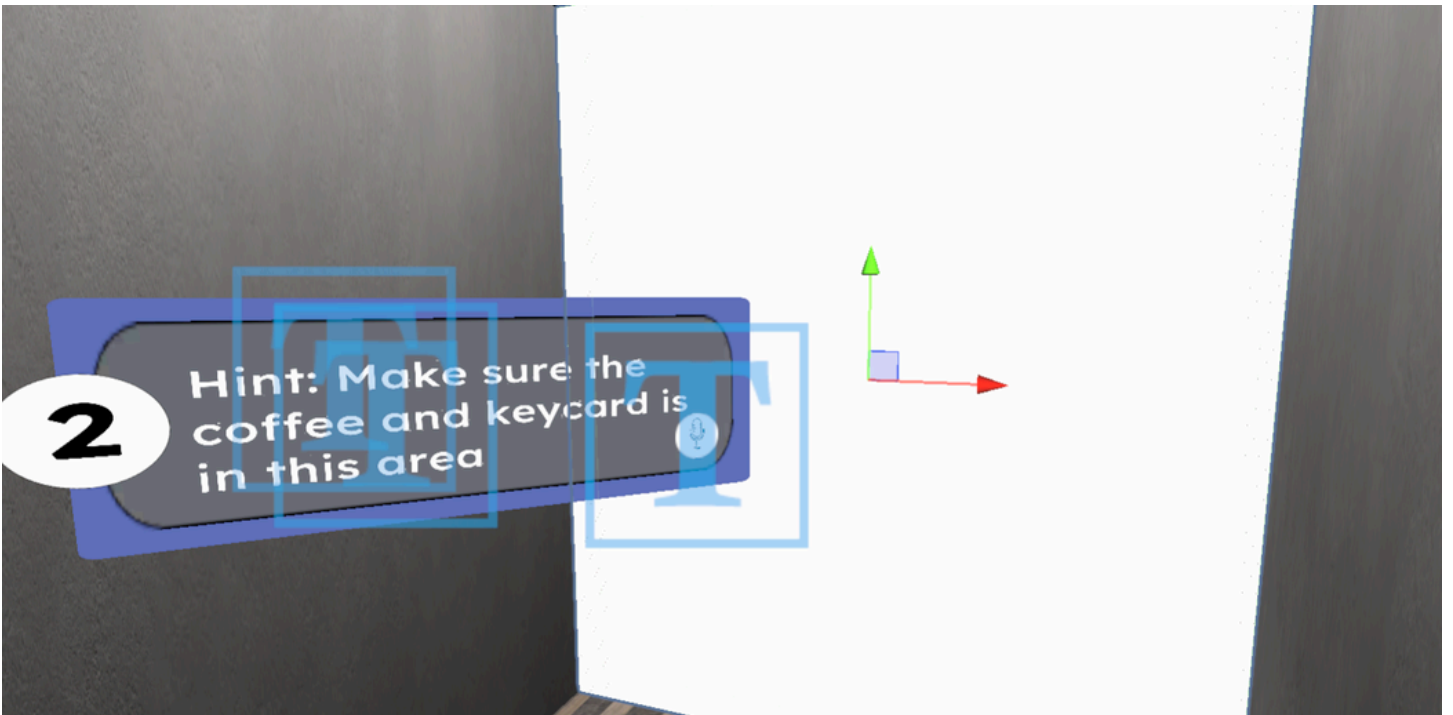


Room 3 Redesign: The two things to redesign for room 3 is the ramp and the doors.

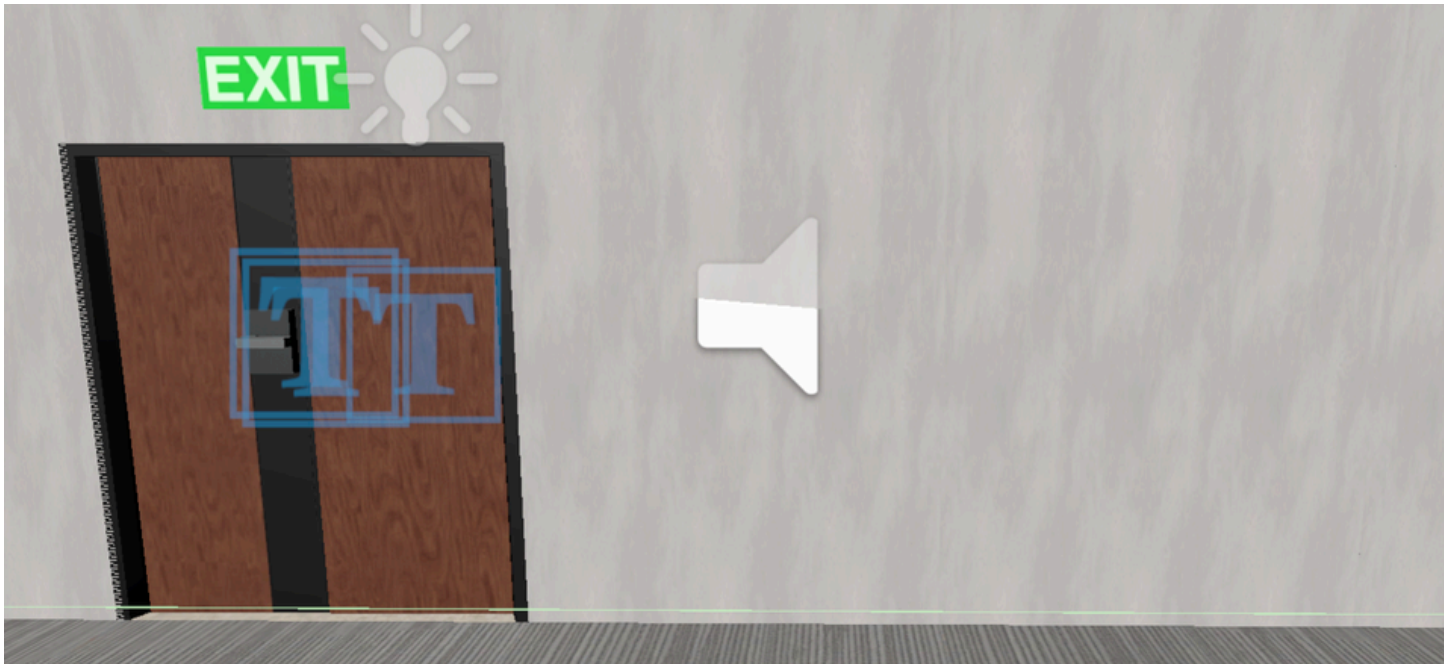




Room 4 Build: For the escape door to appear the player must bring the keycard and coffee mug to the escape door.

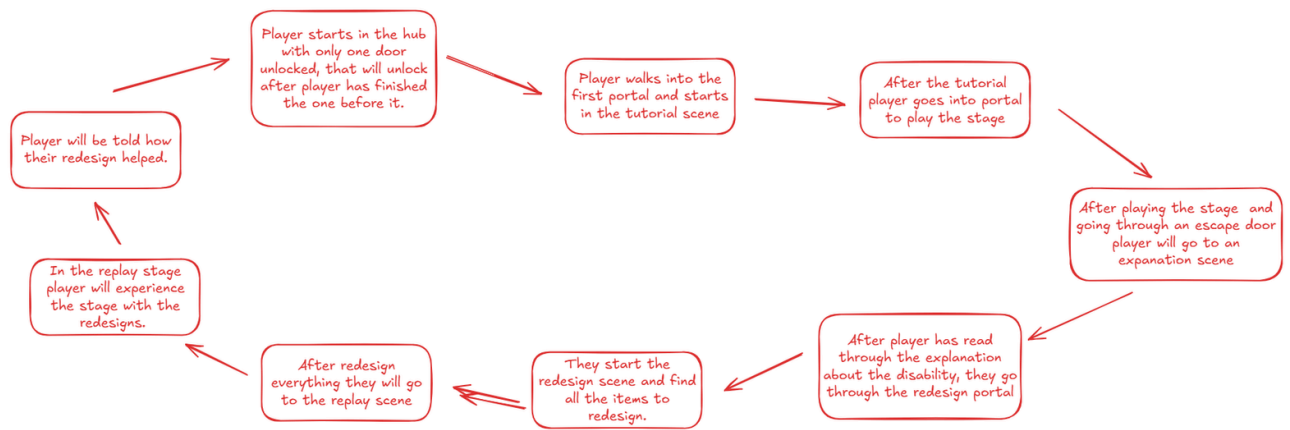


Room 4 Redesign: Player have to redesign the space beside the door and the door itself.



Gameplay:

General Gameplay Loop:



Room 1: Color Vision Deficiency Navigating a Color-Coded Transit Station



- Red–green colour vision deficiency (CVD) was selected as it is the most common form of colour vision deficiency.
- Despite its prevalence, it is often overlooked in public design because:
- Affected individuals can usually function independently.
- The disability is not immediately visible to others.
- Public environments frequently rely on colour-coded systems, assuming colour is universally distinguishable.

Experience Flow Overview

- The user enters the train station environment and is presented with a clear objective, proceed to a specified train line.

- After taking lift down to the train station, the user observes 2 platforms that appear structurally similar and share the same destination name.
- The only distinguishing feature between the platforms is their colour coded signage, which under the simulated colour vision deficiency appears ambiguous and difficult to differentiate.

As the user explores the space:

- Approaching the 2 different platforms does not provide additional clarity.
- Moving within the station does not reveal new information or confirmation cues, even with the map that has been provided.
- The user is forced to make a choice between platforms without sufficient information and time to do so confidently .

Room 2: Short Sightedness – Finding an Office Room in a Dim Corridor



- Short-sightedness (myopia) was chosen because it is:

One of the most common visual conditions worldwide.

Often perceived as a minor or easily solvable issue by the general public.

- Many environments assume users:

Have clear distance vision.

Can read signage from afar without difficulty

Experience Flow Overview

- The user begins in an office corridor and is given a simple navigation goal, locating a specific room.
- As the user moves through the corridor:

Room numbers and directional signs are visible from a distance but remain unreadable due to blur and low contrast.

The user must repeatedly move closer to doors and signs to identify the correct location.

- During navigation:

Obstacles or clutter become noticeable only at close range.

This increases hesitation, backtracking, and minor navigation errors.

Room 3: Wheelchair User in a Public Environment



- Wheelchair users were chosen because:

Mobility barriers are among the most common yet overlooked accessibility issues.

Many environments are labelled as “accessible” but still require excessive effort to navigate.

- Accessibility is often treated as:

A checklist requirement rather than a usability consideration.

Experience Flow Overview

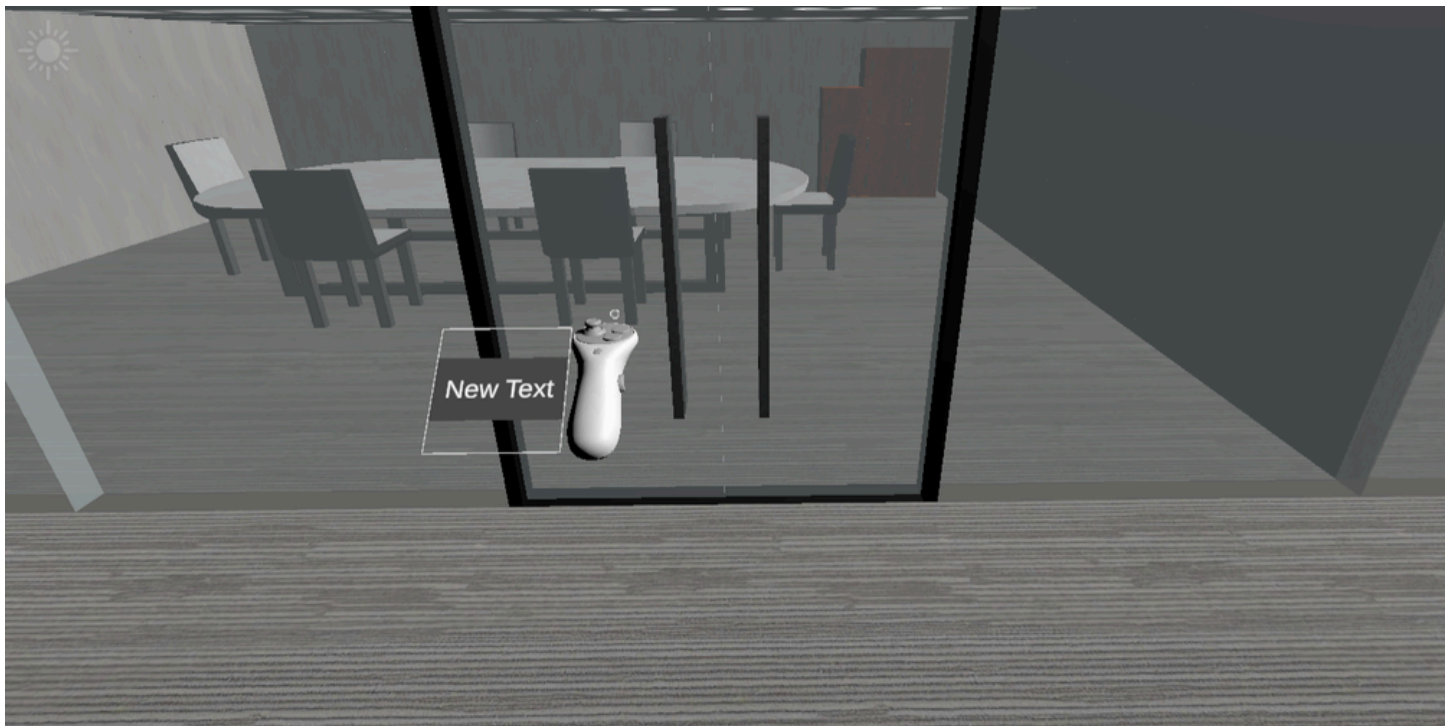
- The user begins outside a public building in a wheelchair.
- The objective is to enter the building and proceed inside.
- Upon approaching the entrance: A ramp is visible, suggesting accessibility.
- As the user ascends the ramp: Movement slows due to the steep incline. Maintaining momentum becomes difficult.
- Upon reaching the top:

The user must pull/push open a door.

Positioning the wheelchair close enough to operate the handle is challenging.

The user may experience: Difficulty maintaining balance. Rolling backward slightly after stopping. Although progress is possible, the task feels: Exhausting. Inefficient. Unsafe.

Room 4: One-Handed Use – Carrying Coffee Through a Door



- A person with only one hand was chosen because:

Limb differences are a permanent physical condition, not a situational limitation.

Many environments are implicitly designed for two handed interaction.

- Unlike temporary one-handed scenarios:

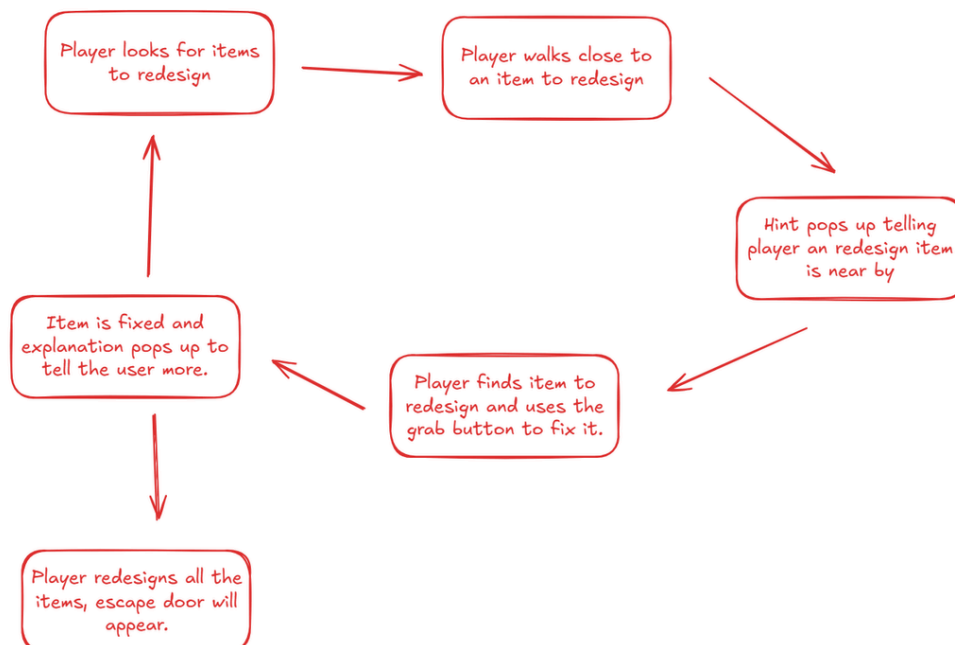
A person with one hand cannot “adapt back” to two handed use.

Environmental barriers are constant rather than occasional

Experience Flow Overview

- The user begins in a office area holding a cup of coffee with their single functional hand.
- The objective is to exit the office while carrying the coffee and keycard whilst not dropping either on the ground.
- As the user approaches the meeting room: An access card reader must be used to unlock the door.
- Because the user has only one hand: They must place the coffee down to use the card reader.
- After authentication: The user must push the door open. The door begins to close automatically if released.
- The user experiences difficulty: Retrieving the coffee while keeping the door open. Preventing the door from closing prematurely.

Room 1 - 4: Redesign Stages



- Because the redesign stages share a consistent gameplay loop across all rooms, they are explained together in this section.
- The goal of including this phase in every room is twofold: to foster empathy by letting players experience these physical barriers firsthand, and to educate them on practical solutions.

- Ultimately, these stages challenge players to actively identify and implement environmental changes that make daily life more accessible for individuals with disabilities.

Redesign Experience Flow Overview :

- The player freely explores the room to locate environmental barriers or objects that require redesigning.
- To prevent frustration and keep the game pacing smooth, a UI hint automatically pops up when the player walks close to an item that can be redesign.
- The player uses the VR **Grab** button to interact with and redesign the object.
- Immediately after the redesign is triggered, an explanation panel appears. This details exactly why the new design is more accessible and beneficial for a person with disabilities.
- Once the redesigns are completed, an exit door spawns, allowing the player to leave and got to the next stage.

Room 1 - 4: Replay Stages

- Similar to the redesign phases, the replay stages share a consistent gameplay loop across all rooms and are therefore grouped together in this section.
- The purpose of this stage is to let the player experience firsthand how their modifications help, preventing them from blindly redesigning items without knowing if the changes are truly beneficial.
- By actively playing through the modified rooms, users can physically test their solutions and feel the practical impact of their designs, rather than just imagining how they might work.

Replay Experience Flow Overview :

- After successfully completing all redesign tasks in the initial rooms, the player is transitioned into the Replay Stage.
- The player must navigate through the exact same sequence of rooms they experienced at the beginning of the game.
- During this playthrough, all the bad design are gone. Every object the player previously had seen is now permanently in its newly redesigned, accessible state.
- By moving through the space a second time, the player directly experiences the stark contrast between the original, frustrating environment and the new, seamless one.
- The player reaches the end of the replayed rooms with much greater ease, reinforcing the educational message of how thoughtful design drastically improves the quality of life for users with disabilities.

References and Credits:

References:

This are the references I used to simulate how it will be like for the people with the disabilities .

Color blindness - [🌐 Red Green Colour Blindness #eyes](#)

Myopia - [🌐 What It's Like to See with -23D Myopia #highmyopia #near-sighted #eyeheath](#)

Wheelchair - [🌐 Design for ramps for wheelchair users](#)

[🌐 WHEELCHAIR MOBILITY Ch13 opening doors](#)

One arm - [🌐 Opening Doors with One Arm/Hand](#)

Credits:

Gemini - [🌐 Google Gemini](#)

Lift Door.cs - Used AI to assist in coding for how the Lift door will open

Redesign Trigger animation- Used AI to make the animation of UI showing up

Wheelchair Scirpts (Grab Points.cs, Velocity supplier.cs and Wheelchair Interactable.cs) - Used AI to help make a script for the wheelchair interaction.

CoPilot - [🌐 Microsoft Copilot: Your AI companion](#)

Used CoPilot for adding comments.